Battle Field Acoustics Group

Mini Rover Build Documentation (WIP)

James Barber

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Contents

1 List of Items

2S1P Ovonic LiPo battery
SparkFun Pro Mini ATmega328-3.3V/8MHz Development Board (Arduino Pro Mini) [1]
WWZMDiB Mini USB to TTL Serial Converter Adapter TTDI Breakout Board [2]
Holybro Kakute H7 Wing Flight Controller [3]
MDD3A Motor Driver [4]
RP1 V2 ExpressLRS 2.4ghz Nano Receiver [5]
Mostop RC Tank Car [6]
HGLRC M100-5883 Compass Module [7]
RadioMaster TX15 [8]

2 Controls

2.1 Manual Mode

To switch to manual mode and arm the rover, turn the S1 dial on the TX15 all the way to the left and switch SA all the way down respectively. Now, the only controls in manual mode are throttle and groundsteering, which are controlled by the right stick.

3 Setting Up Software

3.1 Flashing Arduino Pro Mini

Since the Arduino Pro Mini is built to be as small as possible, it is missing an important chip for flashing the board. The WWZMDiB mini USB to TTL serial converter adapter TTDI breakout board provides this chip. Since this "flashing board" is only used for encoding the Arduino, it is not necessary to include in the final circuit.
In order to connect the flashing board to the Arduino, the boards are connected via the following wiring configuration:

Flashing Board (FB) to Arduino Mini		
FB GND	→	Arduino GND
FB Vcc	→	Arduino Vcc
FB Tx	→	Arduino RXI
FB Rx	→	Arduino TX0
FB DIR	→	Arduino GRN

Once connected, the flashing process is the same as any other Arduino board.

3.1.1 Arduino Pro Mini Software

The .ino code to flash to the Arduino mini is available on GitHub [9]. After selecting the correct board and COM port, go to Tools → Processor and select ATmega328P (3.3V, 8MHz). The build will not work properly otherwise.
The code has built in PWM debugging that can be accessed via a serial monitor in Arduino IDE while the flash board is connected to the Arduino. Using this, pwmMin and pwmMax were determined as 952 and 2008.

3.2 Ardurover

To flash Ardurover to the flight controller, go to Mission Planner → Setup → Install Firmware. As long as the board is connected via USB, the firmware will download to the board.

3.2.1 Parameters

The parameter list for this mini rover is NOT the same as from Basement Creation's video and will result in the rover not working. This list can be found and downloaded on GitHub [9].

3.3 TX15 Config

Go to MDL→Mixes and ensure that the page has, at the least:

Channel	Control
CH1	Ail
CH2	Ele
CH5	SA
CH11	S1

Other channels existing in the mixes page is ok, but for the rover to operate properly these channels must be set as illustrated.

4 Wiring

4.1 Main Circuit Wiring Layout

Kakute H7-Wing Flight Controller (FC)		
FC T6	→	RP1 Rx
FC R6	→	RP1 Tx
FC 5V	→	RP1 5V
FC G	→	RP1 (-)
FC M1	→	Arduino 2
FC M2	→	Arduino 3
FC GPS & I2C-1	→$^{\perp}$	HGLRC GPS
FC 5V	→	GPS 5V
FC TXD	→	GPS RX
FC RXD	→	GPS TX
FC SCL	→	GPS SCL
FC SDA	→	GPS SDA
FC GND	→	GPS GND
⊥ see Sec 4.2		

Flight Controller Power Board (FCPB)		
FCPB BAT-IN	→*	LiPo Battery +
FCPB GND1	→*	LiPo Battery -
FCPB BAT-OUT	→*	MDD3A +
FCPB GND2	→*	MDD3A -
FCPB GND2	→	Capacitor -
FCPB BAT-OUT	→	Capacitor +
* Connected via XT30U-M		

Cytron MDD3A Motor Driver (MDD3A)		
MDD3A VB +	→*	FCPB BAT-OUT
MDD3A VB -	→*	FCPB GND2
MDD3A M1A	→	Motor1 +
MDD3A M1B	→	Motor1 -
MDD3A M2A	→	Motor2 +
MDD3A M2B	→	Motor2 -
MDD3A M1A	→	Arduino 6
MDD3A M1B	→	Arduino 5
MDD3A 5VO	→	Arduino Vcc
MDD3A GND	→	Arduino GND
MDD3A M2A	→	Arduino 10
MDD3A M2B	→	Arduino 11
* Connected via XT30U-M		

Arduino Pro Mini		
Arduino 6	→	MDD3A M1A
Arduino 5	→	MDD3A M1B
Arduino Vcc	→	MDD3A 5VO
Arduino GND	→	MDD3A GND
Arduino 10	→	MDD3A M2A
Arduino 11	→	MDD3A M2B
Arduino 2	→	FC M1
Arduino 3	→	FC M2

4.2 Important Considerations When Wiring

The HGLRC M100-5883 GPS Module comes with a pre-wired connector that fits in the GPS port of the flight controller. <u>If the GPS is wired to the FC in this way and powered, the GPS will be damaged.</u> Instead, intricately follow the wiring layout in the tables in Sec. 4.1.

5 Connecting RadioMaster TX15 to RP1 V2 ExpressLRS 2.4ghz Nano Receiver

1. On TX15 MDL → Model Setup → Internal RF.
    - Ensure Mode is set to CRSF.
    - Make sure Model Setup → External RF Mode is OFF.
2. On TX15 SYS → Tools → ExpressLRS
    - Set packet rate to something with 2.4GHz. 50 2.4G worked best in testing.
    - TX Power → Max Power set to 10mW.
    - WiFi Connectivity → Enable Wifi
    - On computer, connect to ExpressLRS TX network. Password is expresslrs.
    - New tab in internet browser should open automatically. If not, search 10.0.0.1.
    - Enter any binding phrase. Set AirPort UART baud to 420000. Click save and reboot.
    - Turn off TX15.
3. Ensure RP1 is connected properly to FC and power FC via either USB or LiPo.
    - Wait about 60s until RP1 light is flashing very fast. This means RP1 WiFi network is running.
    - On computer, connect to ExpressLRS RX network. Password is expresslrs.
    - Open 10.0.0.1. again.
    - Enter same binding phrase, set UART baud to 420000, and set mode or whatever to CRSF.
4. Turn on TX15 and go to SYS → Tools → ExpressLRS → Bind. The TX15 and RP1 should automatically connect. To show a successful connection, the light on the RP1 stays static.

6 Troubleshooting

Useful Resources

[1] *Arduino Pro Mini 328 - 3.3V/8MHz*. en. URL: `https://www.sparkfun.com/arduino-pro-mini-328-3-3v-8mhz.html` (visited on 06/15/2026).

[2] *Amazon.com: WWZMDiB Mini USB to TTL Serial Converter Adapter, 3.3V/5.5V, FTDI Breakout Board for Arduino : Electronics*. URL: `https://www.amazon.com/WWZMDiB-FT232RL-Converter-Adapter-Breakout/dp/B0BJKCSZZW` (visited on 06/15/2026).

[3] *Holybro KakuteH743-WING — Copter documentation*. URL: `https://ardupilot.org/copter/docs/common-kakuteh7wing.html` (visited on 06/07/2026).

[4] *PN00218-CYT12 3Amp 4V-16V DC Motor Driver (2 Channels) MDD3A*. en. URL: `https://makermotor.com/pn00218-cyt12-3amp-4v-16v-dc-motor-driver-2-channels-mdd3a/` (visited on 06/09/2026).

[5] *RP1 V2 ExpressLRS 2.4ghz Nano Receiver*. en. URL: `https://radiomasterrc.com/products/rp1-expresslrs-2-4ghz-nano-receiver` (visited on 06/07/2026).

[6] *Amazon.com: Mostop RC Tank Car, 1:16 Scale 2.4Ghz Remote Control Crawler High Speed Tank, Off-Road RC Tracked Vehicles 360°Rotating Drifting Car with 2 Batteries, Military Truck for Kids Adults : Toys & Games*. URL: `https://www.amazon.com/Mostop-Transport-Military-Batteries-360%C2%B0Rotating/dp/B0DBTL5LRR?th=1` (visited on 06/15/2026).

[7] HGLRC. *HGLRC M100-5883 GPS*. en. URL: `https://www.hglrc.com/products/m100-5883-gps` (visited on 06/09/2026).

[8] *User Manuals*. en. URL: `https://radiomasterrc.com/pages/user-manuals` (visited on 06/15/2026).

[9] James Barber. *General · JamesB-C/Mini_Rover*. en. URL: `https://github.com/JamesB-C/Mini_Rover`.